

04.03.26

HTTP → web based

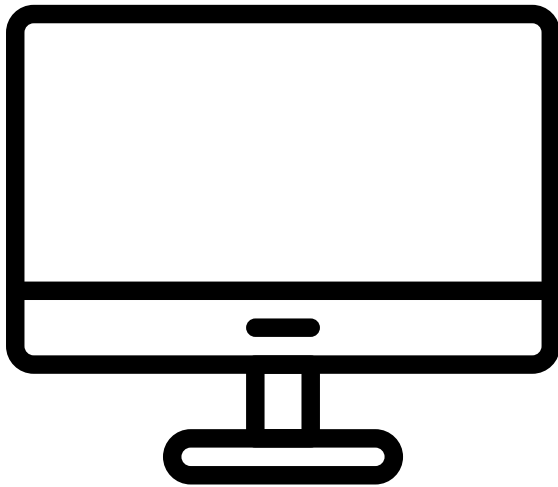
COMPUTER NETWORKS

CSE 321

Let's send email

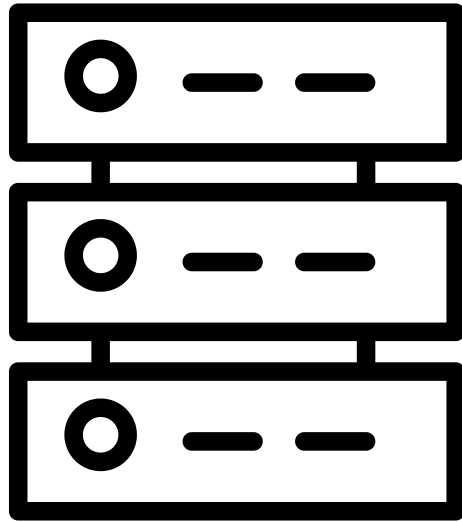
E-mail Model

২ টি client থাকবে



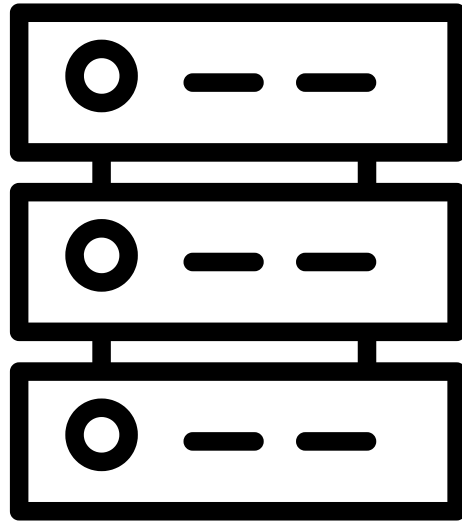
Sender (User)

client



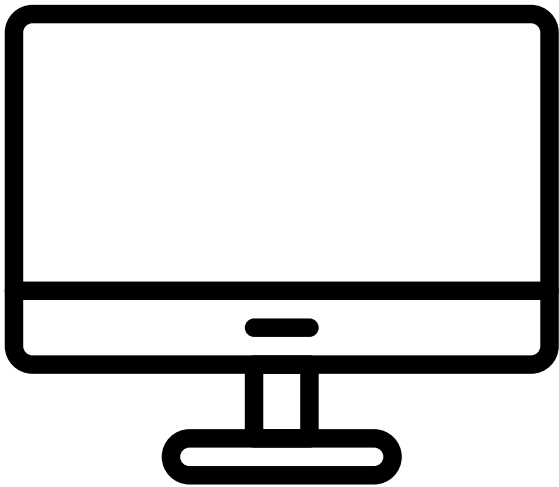
Mail Server (Sender গা)

e.g. gmail



Mail Server (Receiver গা)

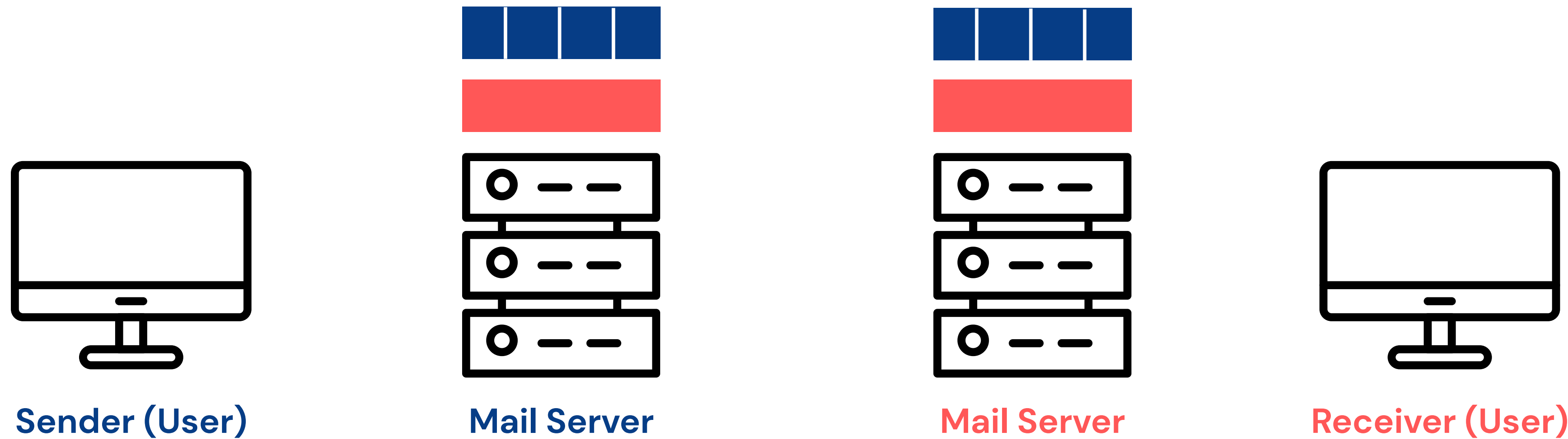
e.g. yahoo



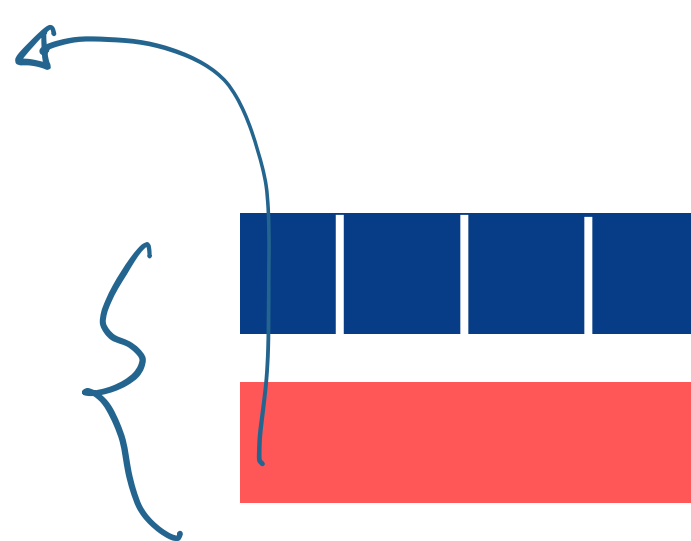
Receiver (User)

client

E-mail Model



user এর কাছে আসা
mail দু'লা রাখে
২ টা main
component

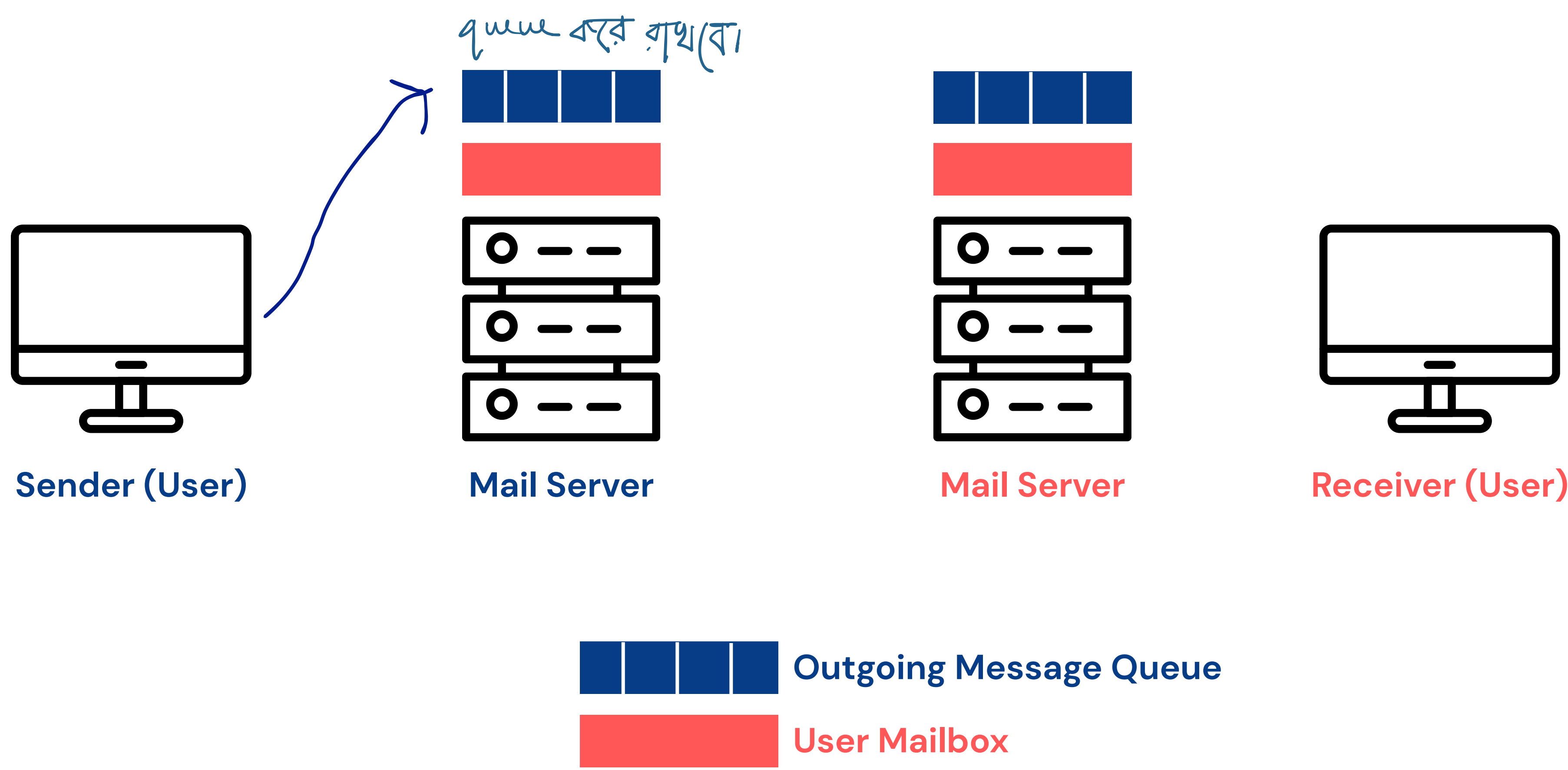


mail টা পাঠানোর সময় server down.
queue ত রাখে try
করে। successful
হলে queue থেকে
remove

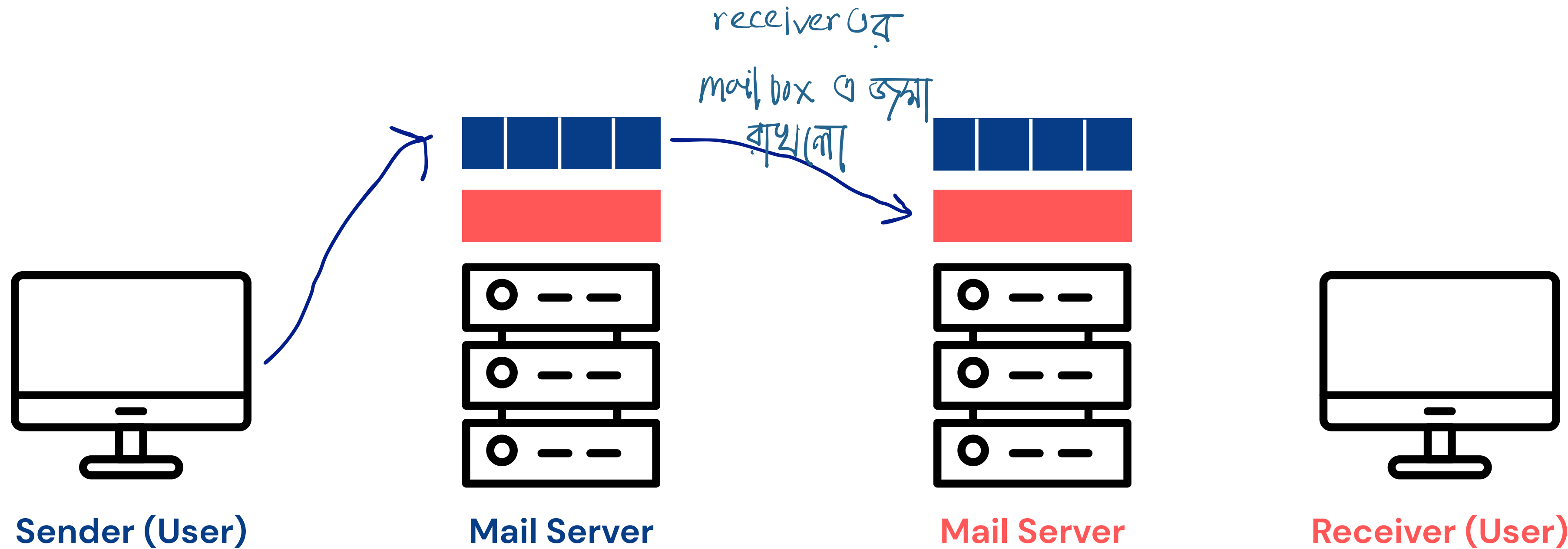
Outgoing Message Queue

User Mailbox (per user ১টা থাকে)

E-mail Model



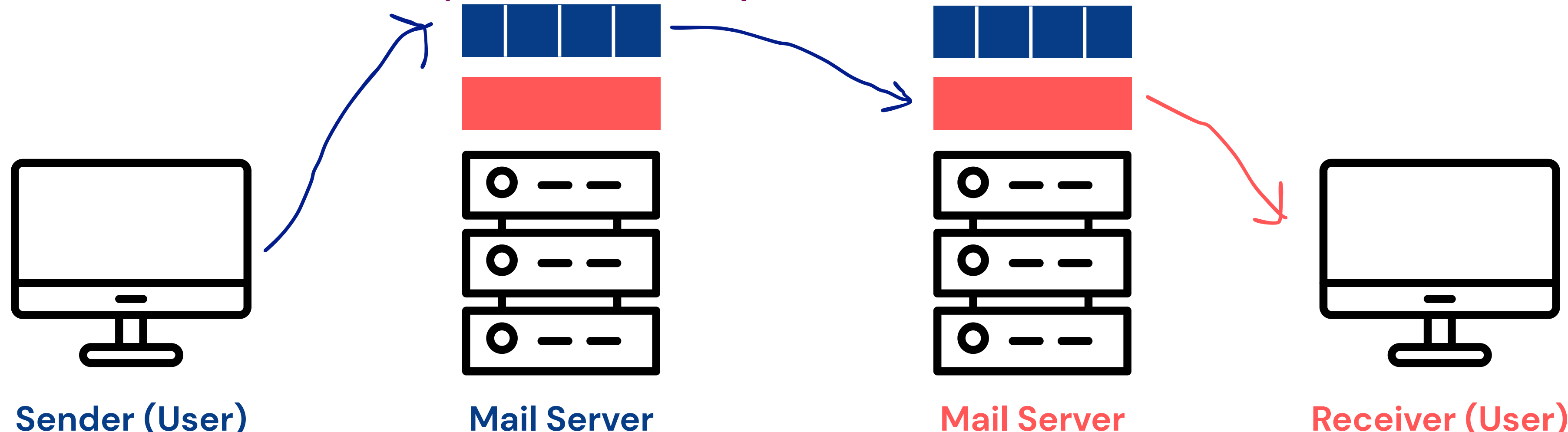
E-mail Model



E-mail Model

HTTP এর basic philosophy এর সাথে outgoing message queue এর diff.

client request পাঠায়
server response দেয়



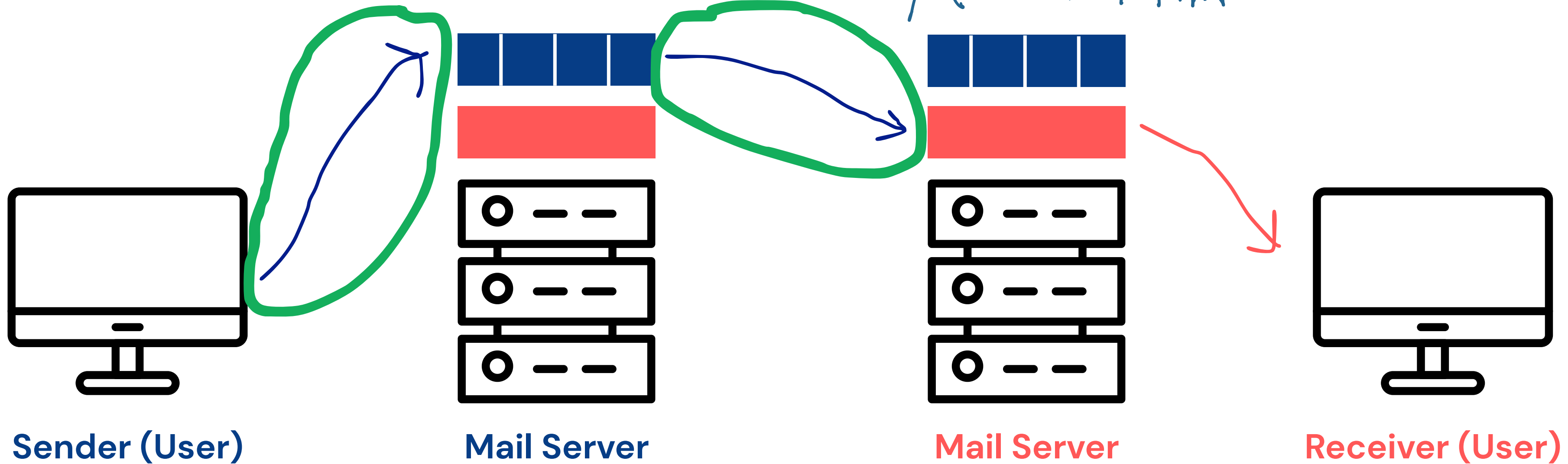
 Outgoing Message Queue
 User Mailbox

Fundamental Difference with HTTP

HTTP → **Client pull**

Here → **Client push**

→ server কে বলছি এটা পাঠাও । server কে push করছে ।
(post req দিয়ে implement করা হবে না কেন?)
→ body তে mail পাঠানো হবে



SMTP

Simple Mail Transfer Protocol

① client — push করে

PULL করা লাগলে $\rightarrow$ not effective

$\rightarrow$ যতগুলো mail server আছে সব্বার কাছে একটা পাবার request পাঠাতে।

SMTP

Simple Mail Transfer Protocol

- **Push Protocol (Otherwise recipient server had to ask every servers if they got any message for it constantly)**

→ not efficient

SMTP

Simple Mail Transfer Protocol

- Push Protocol (Otherwise recipient server had to ask every servers if they got any message for it constantly)
- Built for asynchronous communication. It can queue messages and retry multiple times if failed.

যদি server down থাকে SMTP queue তে রাখে কিছুক্ষণ পরপর try করতে থাকে। নিজেই retry করে। যদি mail address exists হৈ না করে তখন sender কে notify করে।

HTTP তে retry ও client এর
করা লাগে। Notify করে Please
retry!

SMTP

Simple Mail Transfer Protocol

- Push Protocol (Otherwise recipient server had to ask every servers if they got any message for it constantly)
- Built for asynchronous communication. It can queue messages and retry multiple times if failed.
- SMTP predates HTTP. Most of servers were already using SMTP.

Web page send - receive আরে পরে এসেছে। SMTP
HTTP আরার আগেই standardize হয়ে গিয়েছে।

SMTP

Simple Mail Transfer Protocol

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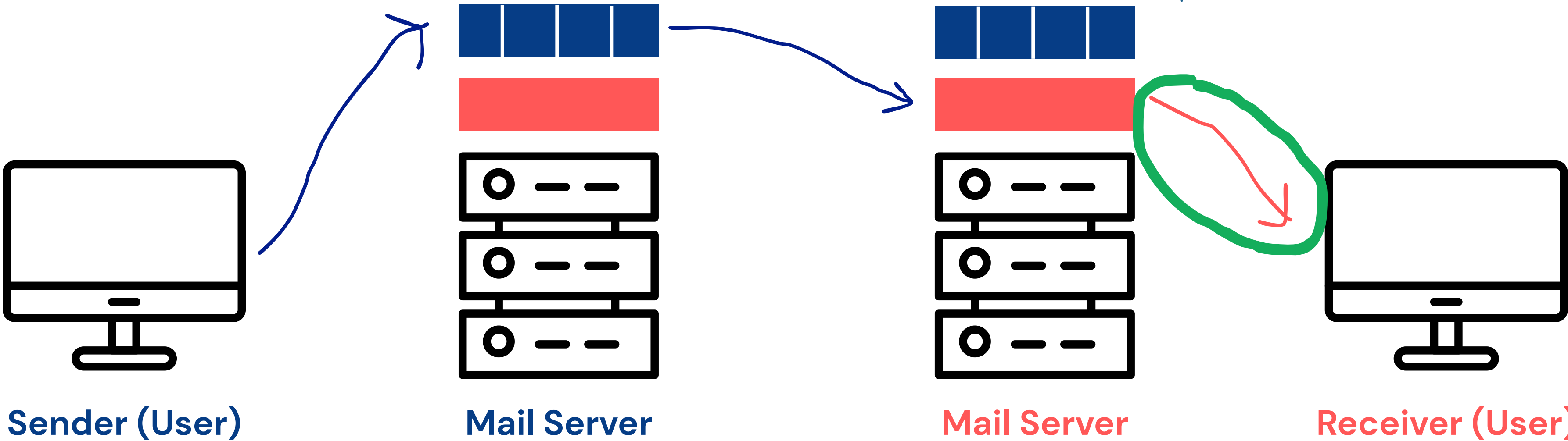
→ Mail server always → SMTP

Nowadays, communicating mail server from web browsers uses HTTP/HTTPS

browser এর মাধ্যমে (webmail use করেন) → HTTP/HTTPS
use হয়

Accessing Emails

Red part → pull করা লাগে।
SMTP use করতে পারিনা।
HTTP ও না।



Accessing Emails

IMAP (Internet Message Access Protocol)

- Synchronization between multiple devices
- Selective downloading
- Manages folders, flags etc

mobile, laptop

HTTP দিয়ে করা
যেতে but complexity
বাড়বে।

প্রথমে শুধু header (sender, subject, time)
click করলে full body.

↓
maintain করতে সফট লোড।

cookie দিয়ে করা যেতে। But কতগুলো cookie রাখা।

Accessing Emails

IMAP (Internet Message Access Protocol)

- Synchronization between multiple devices
- Selective downloading
- Manages folders, flags etc

→ both way communication (HTTPও)

but HTTP response code ফিল্ড fixed. IMAP দিয়ে easily synchronize করা যায়।

কিছু স্টেপ HTTP দিয়ে রানও
main কাজ SMTP দিয়ে হয়।

POP3 (Post Office Protocol v3)

- Good for single device access → dedicate device দিয়ে mail দেখা বা
- Deletes from server after download

→ mail device এ download (read করা রানও)
server এ আর রাখা যোনা, so অন্য device
দিয়ে দেখতে চাইলে পাওয়া না।

SMTP → uses TCP

(not in syllabus)



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